CE 310

Week 8 — Review Session

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Today

- Walk through Weeks 1–7 one at a time: **core idea** → **worked number** → **common mistake**
- Nothing to turn in — ask questions any time
- Full detail (formula sheet, all worked examples) is in the **Study Guide** (`Wk08.02`) — this is the highlight reel, not a replacement
- Wed = Concept Exam (closed computer) · Fri = Lab Practicum (open Excel)

Week 1 — The Flaw of Averages

Using the **average** to represent a whole distribution breaks down whenever the input–output relationship is nonlinear. For equipment sizing, the **peak** matters more than the mean.

Worked number. Tucson outdoor temp: $\mu = 68.75°F$, MAX $\approx 106.9°F$. A chiller sized for 68.75°F fails on every day above its rated capacity.

> Common mistake: reporting only the mean and calling the system "adequate on average." Average-adequate ≠ peak-adequate — always check the tail.

Week 2 — Spread and Shape

STDEV, SKEW, and IQR describe how spread out and asymmetric a distribution is. Histograms/box plots reveal shape that a single number hides.

Worked number. Cooling_kWh in Tucson: $\sigma = 25.3$ kWh, positive skew (long right tail — a few peak-cooling hours dwarf the median).

> Common mistake: STDEV (sample, $\div(n-1)$) vs. STDEVP (population, $\div n$). With engineering samples, use STDEV.

Week 3 — Conditional Probability

P(A | B) = "given B happened, what's the chance A also happened?" = joint count ÷ conditioning count.

Worked number. P(BusinessHours="Yes" | Season="Summer") = COUNTIFS(Season,"Summer",BH,"Yes") / COUNTIF(Season,"Summer").

> Common mistake: dividing by the full dataset (8,760) instead of the conditioning count. P(A|B) denominates on B, not on everything.

Week 4 — Poisson Distribution

Models the count of discrete, independent events in a fixed interval, given rate λ.

Worked number. $\lambda = 3$ failures/month → $P(X=0)$ = POISSON.DIST(0,3,FALSE) = $e^{-3} \approx 0.0498$ (≈5% chance of zero failures).

> Common mistake: using λ for the wrong time window. Rate = 6/hr but need a 15-min probability? Convert first: $\lambda = 6/4 = 1.5$.

Week 5 — Normal Distribution & Z-Scores

Normal = symmetric bell curve; $z = (x - \mu) / \sigma$ standardizes any Normal variable. 68–95–99.7 rule for $\pm1\sigma/\pm2\sigma/\pm3\sigma$.

Worked number. Beta concrete: μ=4,367 psi, σ=484 psi → P(fc < 4,000) = NORM.DIST(4000,4367,484,TRUE) ≈ 22.4%. Beta fails ACI 318 (max 10% below spec). Alpha: NORM.DIST(4000,4632,344,TRUE) ≈ 3.3% — compliant.

> Common mistake: using NORM.DIST(...,FALSE) — that's density, not probability. Use TRUE (cumulative) for a probability.

Week 6 — Empirical Percentiles & Exceedance

LARGE/PERCENTILE compute percentiles directly from ranked data — no distributional assumption needed. Exceedance = P(X > threshold).

Worked number. Traffic speed (500 rows): NORM.INV(0.85,54.4,7.8) = 62.5 mph (Normal model 85th %ile) vs. LARGE(Speed_mph,75) from ranked data (empirical 85th %ile; 15% × 500 = 75). Models agree near the 85th percentile; gaps widen at extreme tails.

> Common mistake: expecting PERCENTILE(data,0.85) and LARGE(data,75) to match exactly in a 500-row dataset. PERCENTILE interpolates; LARGE returns an actual data value — close, not identical.

Week 7 — Correlation and Covariance

Pearson r (–1 to 1) measures linear relationship strength/direction; r^2 = fraction of variance explained. Covariance has units, r doesn't.

Worked number. r(Outdoor_Temp_F, Cooling_kWh) = 0.7752 all-hours vs. 0.9539 business-hours-only — stronger during business hours because the HVAC system runs at full, predictable capacity.

> Common mistake: reading $r \approx 0$ as "no relationship." It only rules out a LINEAR one — a strong U-shaped pattern can still have $r \approx 0$.

Questions?

Study Guide (`Wk08.02`) has the full formula sheet and every worked example.

Wed: Concept Exam · Fri: Lab Practicum

You've done this work seven times already. Trust it.